

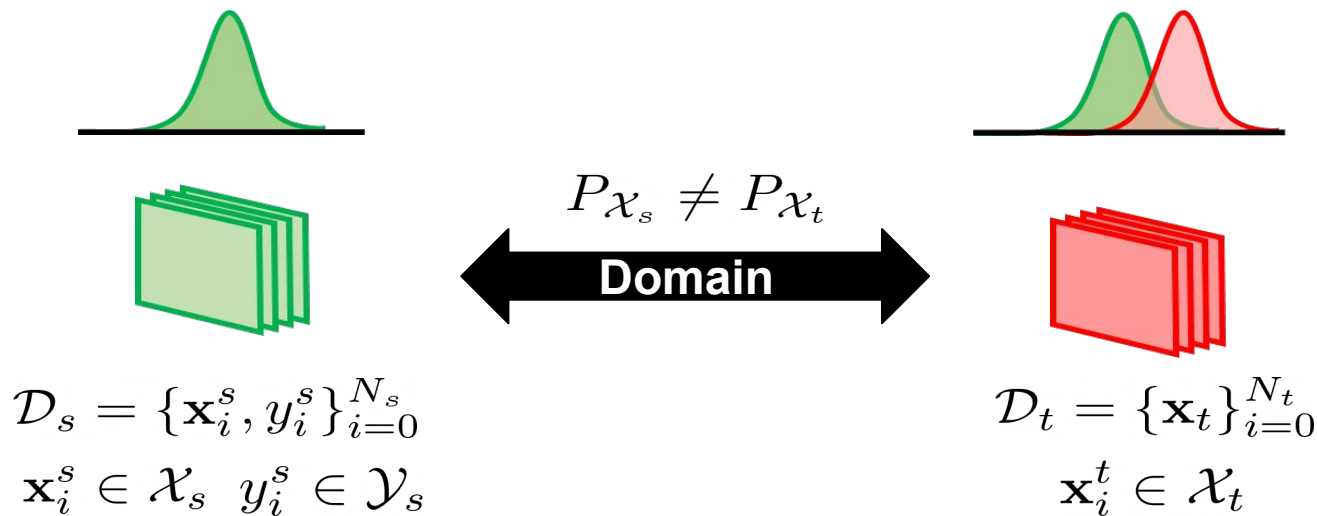
NC-TTT: A Noise Contrastive Approach for Test-Time Training

David Osowiechi*, Gustavo A. Vargas Hakim*, Mehrdad Noori, Milad Cheraghali, Ali Bahri, Moslem Yazdanpanah, Ismail Ben Ayed, Christian Desrosiers

*equal contribution

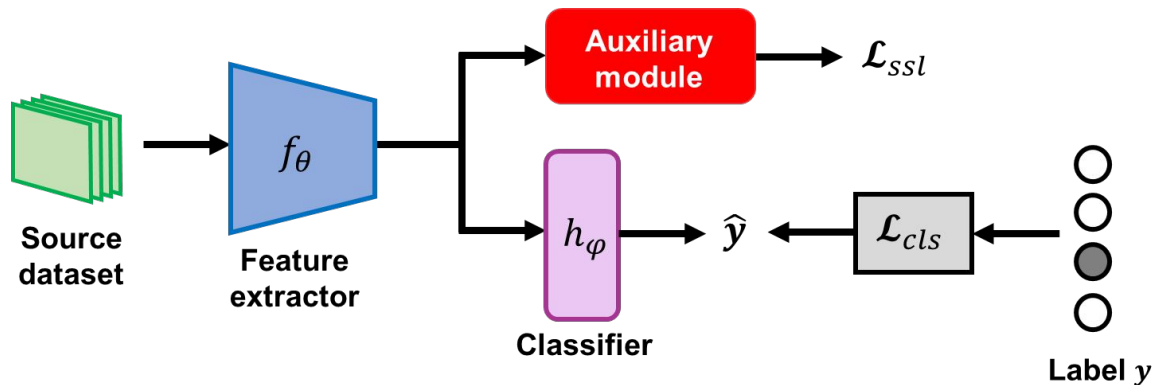
Test-Time Adaptation

In Test-Time Adaptation, a model is adapted to a target dataset without accessing the training (source) data:



Test-Time Training

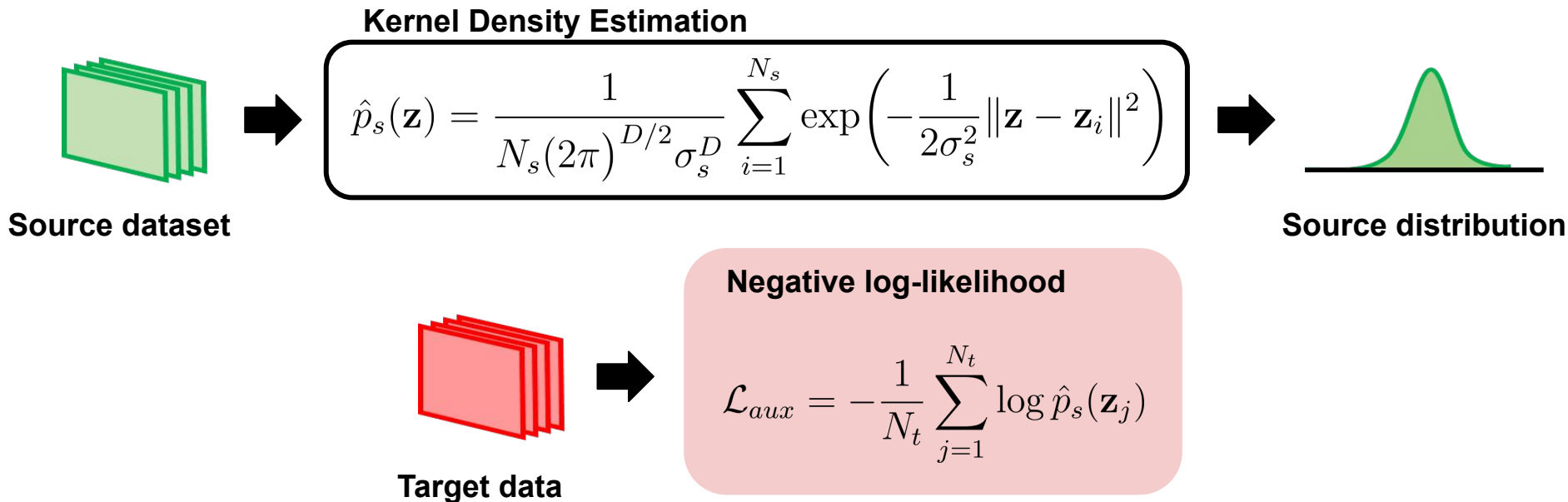
An auxiliary task is used at training time and at test-time:



Examples: TTT (rotation prediction), TTT++ (contrastive learning), TTTFlow (normalizing flow), ClusT3 (Mutual Information clustering)

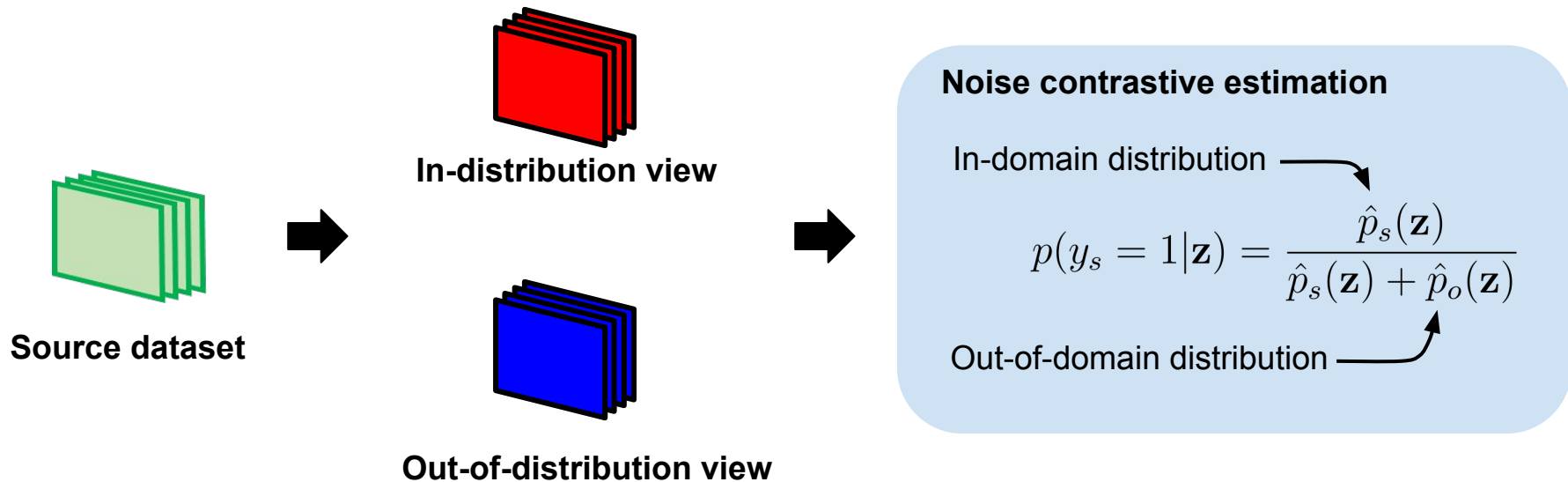
Modelling source distribution

We could measure the likelihood of a target sample with respect to the source distribution.



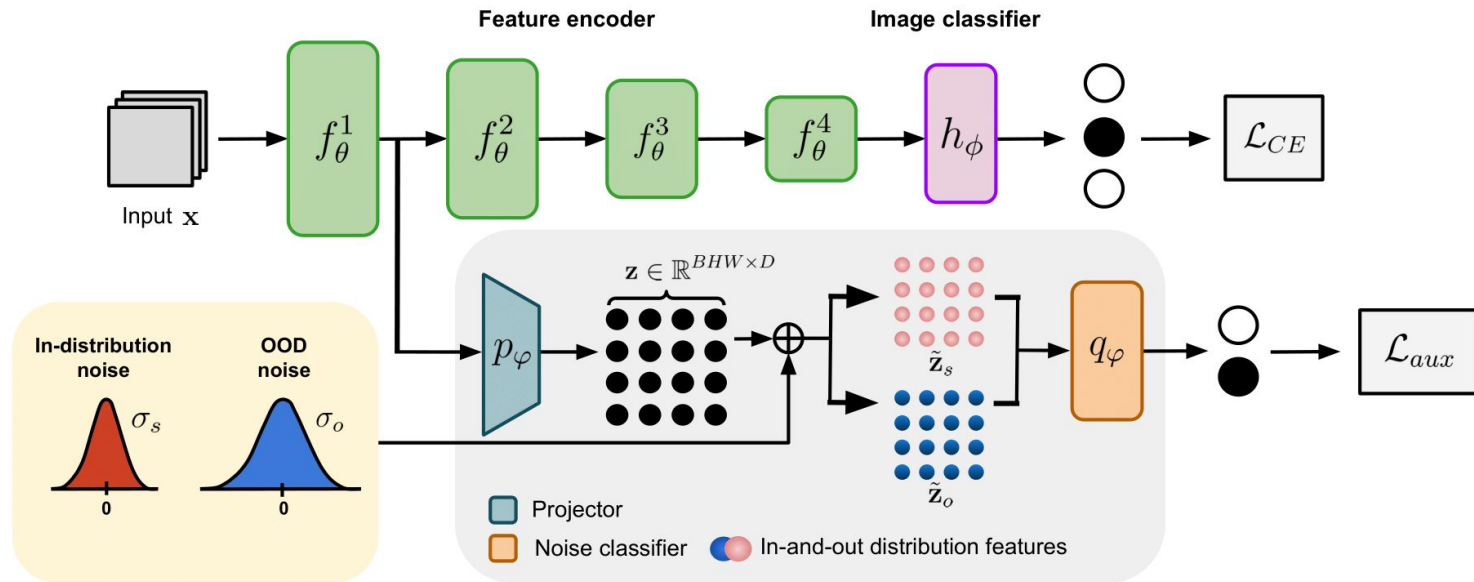
Modelling source distribution

But we have no access to source data at test-time. We then use Noise Contrastive Estimation to model the source distribution during training:



NC-TTT: architecture

A discriminator learns to classify two noisy views of source feature maps. We model in and out distribution using Gaussian noise with equal mean and different variance:



NC-TTT: pseudo-labelling

The noise samples overlap, which would confuse the discriminator. We then assign pseudo-labels as $p(y_s = 1 | \tilde{\mathbf{z}}) = \text{sigmoid}(u)$ and using:

$$u = \frac{1}{2} \left(\frac{1}{\sigma_o^2} - \frac{1}{\sigma_s^2} \right) \|\epsilon_{i,m}\|^2 + D \log \left(\frac{\sigma_o}{\sigma_s} \right)$$

$\epsilon_{i,m} = \tilde{\mathbf{z}}_{i,m} - \mathbf{z}_i$ is the difference between
a feature map and its noisy view

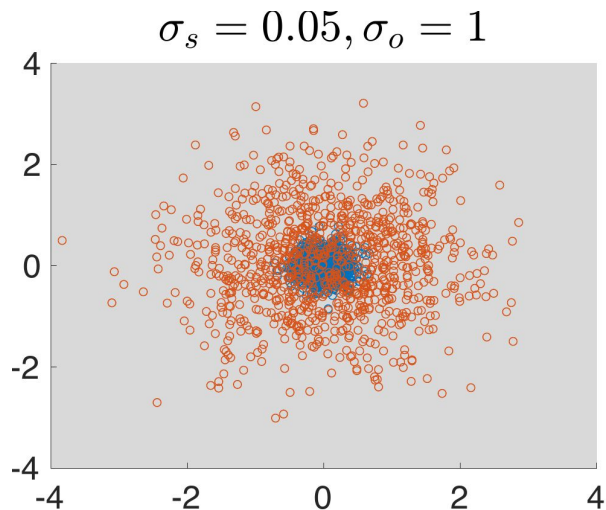
Final auxiliary loss

Using $\tilde{p}_{i,m} = p(y_s = 1 | \tilde{\mathbf{z}}_{i,m})$ as soft label, the discriminator is trained with:

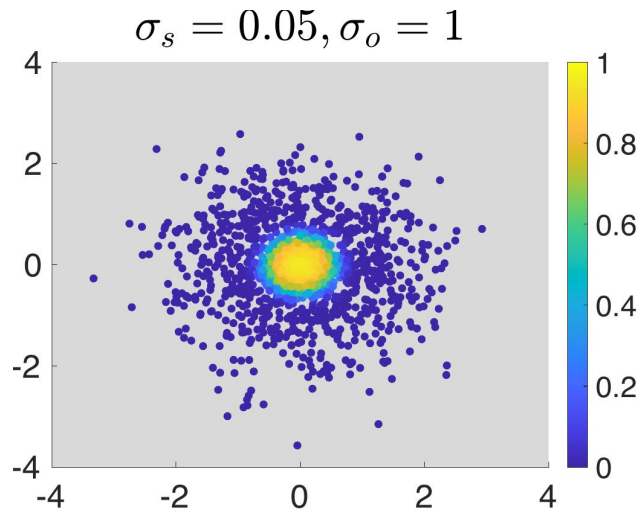
$$\mathcal{L}_{aux} = -\frac{1}{2MN_s} \sum_{i=1}^{N_s} \sum_{m=1}^{2M} \tilde{p}_{i,m} \log q_{\varphi}(\tilde{\mathbf{z}}_{i,m}) + (1 - \tilde{p}_{i,m}) \log (1 - q_{\varphi}(\tilde{\mathbf{z}}_{i,m}))$$

NC-TTT: pseudo-labelling

Each noisy sample is assigned a soft pseudolabel:



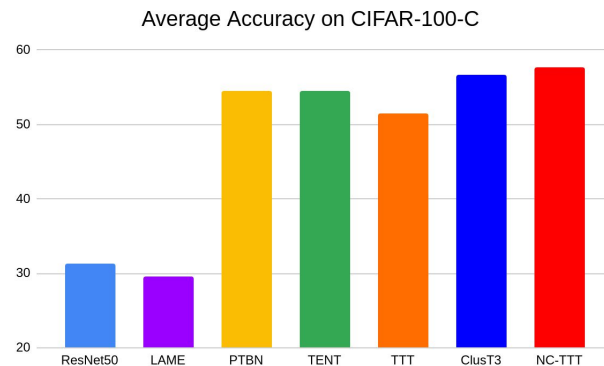
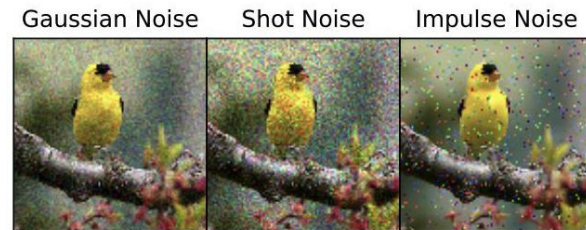
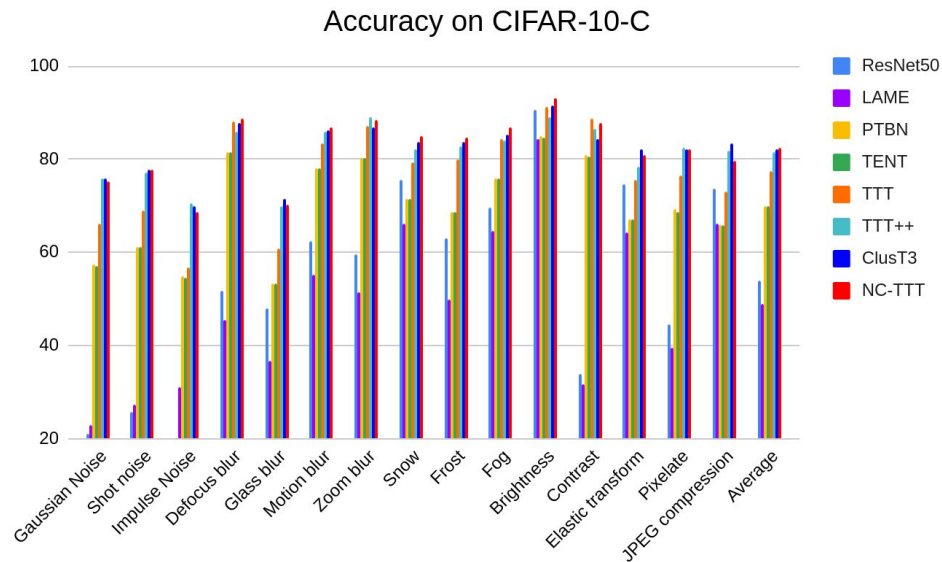
Original noisy samples



Probabilistic pseudolabels

Results: common corruptions

Using CIFAR-10/100-C: 15 different corruptions with 10K samples per corruption



Results: sym-to-real

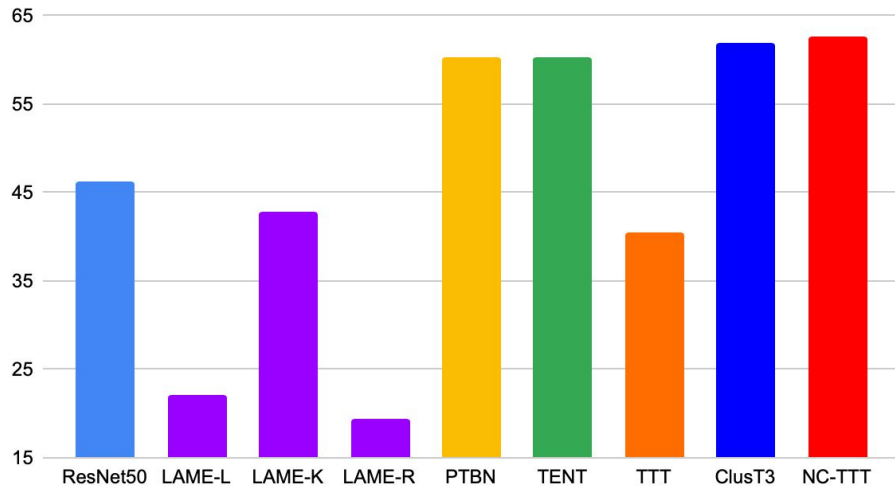
Using VisDA-C: 152,397 3D renderings → 72,372 video frames



bus

car

Accuracy on VisDa-C



Future work

- Exploring different types of noise within the same framework.
- Open question in TTT: what are the mechanisms that help improving a model at test-time?



Contact us if you have any questions:

david.osowiechi.1@ens.etsmtl.ca

gustavo-adolfo.vargas-hakim.1@ens.etsmtl.ca

And look for the code and the whole article here:

<https://github.com/GustavoVargasHakim/NCTTT>

